

Graduate Systems

Assingment 1

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Part A

Program A - I used a loop to create multiple processes using fork() in parent process

Program B - I used a loop to create multiple Threads using pthread_create() in parent process

Part B

I created three worker functions:

- cpu() - this is a function which will be more cpu intensive
- io() - this is a function which will be more io intensive
- mem()- this is a function which will be more mem intensive

Part C

In this part we analysed different config for above program program A + cpu, program b + cpu , program A + mem , program B + mem, etc

And results and stored in MT25083_PART_C_CSV.csv file in the folder and table is as shown below

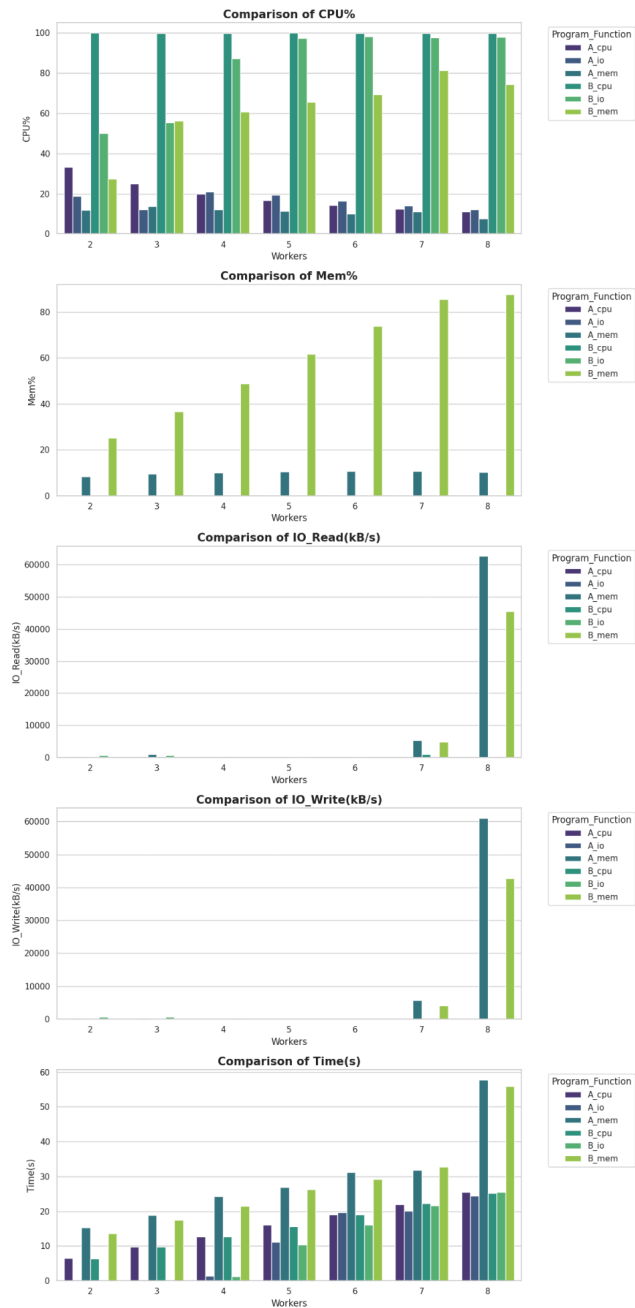
Program+Function	CPU%	Mem%	IO_Read(kB/s)	IO_Write(kB/s)	Time(s)
MT25083_Part_A_Program_A_cpu	16.609	0.1	4.63663	5.66698	16.22245745
MT25083_Part_A_Program_A_io	19.2472	0.1	12.1527	7.76554	10.39067555
MT25083_Part_A_Program_A_mem	13.1008	10.3352	6.26254	4.04365	24.50120583
MT25083_Part_A_Program_B_cpu	99.908	0.1	6.48847	3.94954	25.60561887
MT25083_Part_A_Program_B_io	97.7415	0.1	3.74275	3.79595	25.11363825
MT25083_Part_A_Program_B_mem	56.0835	85.354	27738.9	30848.3	68.01986809

Here we can see that program A + cpu and program A + io are both heavy on cpu but low on mem and mem in both case is high while using thread we can also see that there is more cpu compute.

Part D

In part D we can see results in MT25083_PART_D_CSV.csv and also the plots made for D part is given below.

Performance Factor Comparison: Program A vs Program B



In this also we can see that amount of cpu used by threads are way higher like they are using 100% of that core. IO show results at a higher worker count. Mem in A program is nearly constant while increasing the count of processes but when done same in threads we see that there is increase in mem usage.

AI usage:

- 1) I have used ai to create the bash file
- 2) I have used ai to debug some parts of my code in workers where i had to use cpu, io and mem intensiv function
- 3) i used ai to create Makefile
- 4) i used AI to create readme file

<https://gemini.google.com/share/2c5dca968bb1>

Github: https://github.com/sarvagya334/GRS_Assingment_MT25083/tree/main/GRS_PA01